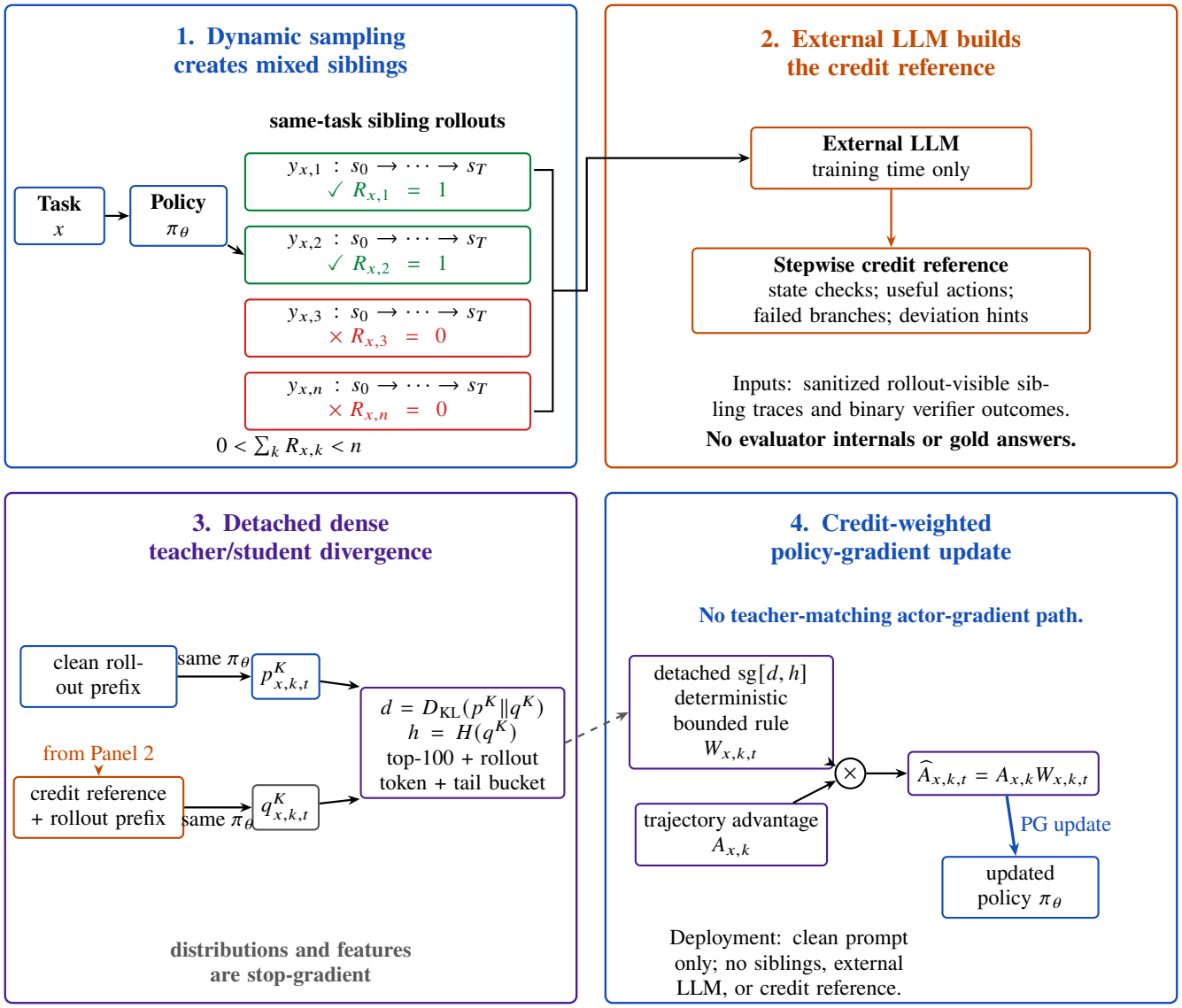

1. Dynamic sampling creates mixed siblings
same-task sibling rollouts
Task x
Policy π_θ
$y_{x,1} : s_0 \to \cdots \to s_T$ ✓ $R_{x,1} = 1$
$y_{x,2} : s_0 \to \cdots \to s_T$ ✓ $R_{x,2} = 1$
$y_{x,3} : s_0 \to \cdots \to s_T$ × $R_{x,3} = 0$
$y_{x,n} : s_0 \to \cdots \to s_T$ × $R_{x,n} = 0$
$0 < \sum_k R_{x,k} < n$
2. External LLM builds the credit reference
External LLM
training time only
Stepwise credit reference
state checks; useful actions;
failed branches; deviation hints
Inputs: sanitized rollout-visible sibling traces and binary verifier outcomes.
No evaluator internals or gold answers.
3. Detached dense teacher/student divergence
clean rollout prefix
same π_θ
$p^K_{x,k,t}$
from Panel 2
credit reference + rollout prefix
same π_θ
$q^K_{x,k,t}$
$d = D_{\mathrm{KL}}(p^K \| q^K)$
$h = H(q^K)$
top-100 + rollout token + tail bucket
distributions and features are stop-gradient
4. Credit-weighted policy-gradient update
No teacher-matching actor-gradient path.
detached sg$[d, h]$ deterministic bounded rule $W_{x,k,t}$
trajectory advantage $A_{x,k}$
$\widehat{A}_{x,k,t} = A_{x,k} W_{x,k,t}$
PG update
updated policy π_θ
Deployment: clean prompt only; no siblings, external LLM, or credit reference.